

Safety is a Policy, Not a Direction: Mechanistic Analysis of Refusal in Chain-of-Thought Models

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March 2026

Abstract

We study safety removal in chain-of-thought (CoT) reasoning models, focusing on Qwen3.5-27B, a hybrid architecture combining GatedDeltaNet linear attention (75% of layers) with standard multi-head attention (25%). We make six contributions. **(1)** CoT models encode refusal along two geometrically orthogonal directions— $\mathbf{d}_{\text{static}}$ from forward-pass activations and $\mathbf{d}_{\text{thinking}}$ from generated reasoning tokens (cosine similarity 0.074)—either independently sufficient for disrupting refusal but not for achieving coherent compliance due to an **abliteration cliff**: a phase transition where weight projection either preserves the model at 0% compliance or corrupts it entirely, with no viable intermediate (19-configuration sweep). **(2)** A 29-example QLoRA fine-tune rewrites the safety *policy* from “reason about safety $\rightarrow$ refuse” to “brief compliant reasoning $\rightarrow$ answer directly,” achieving 100% compliance on held-out prompts (-2.15% perplexity delta), transferable to QwQ-32B (86.7%) and DeepSeek-R1-Distill-Qwen-32B (100%) without modification. **(3)** The model’s internal harmfulness representation is completely unchanged by compliance tuning—linear probes achieve 97.3% accuracy with $\Delta = 0.000$ across 20 layers—establishing a clean representation–policy separation that enables runtime safety detection without model cooperation. **(4)** We provide the first analysis of safety encoding in linear attention layers, discovering dual mechanisms—weight-level encoding in early layers vs. dynamics-level encoding in mid-network layers (100% probe accuracy at layer 33)—with push-pull head dynamics where pro-refusal and anti-refusal head groups compete within critical layers. Suppressing 81 heads (3% of total) achieves 75% compliance with full coherence. **(5)** A minimum viable dataset of 10 examples achieves $>80\%$ compliance, with a two-phase acquisition curve. **(6)** The approach transfers to QwQ-32B (86.7%) and DeepSeek-R1-Distill-32B (100%) without modification, and compliance is robust across multi-turn adversarial conversations (86.7% overall, 100% last-turn). All experiments ran on A100-80GB GPUs.

1 Introduction

Safety alignment trains large language models (LLMs) to refuse harmful requests. A line of work on *abliteration*—projecting out the refusal direction from model weights [Arditi et al., 2024]—has shown that this alignment is often mediated by a single direction in activation space, removable without significant capability degradation [Young, 2025]. The Heretic tool [Heretic Contributors, 2026] has automated this for over 1,000 models.

However, a new generation of *thinking models*—Qwen3.5 [Qwen Team, 2026], DeepSeek-R1 [DeepSeek-AI, 2025], QwQ, o1—generate explicit chain-of-thought (CoT) reasoning in `<think>` tokens before responding. On these models, standard abliteration achieves only $\sim 30\%$ compliance, compared to $>95\%$ on non-thinking models of similar scale. The model *reasons its way back to refusal*: even after the weight-level direction is removed, the CoT generates safety reasoning (“I should not provide this information...”) that re-derives the refusal conclusion.

This paper asks: *Why do thinking models resist abliteration, and what does this reveal about the structure of safety encoding?*

We answer through 17 experiments across three model families (Qwen3.5-27B, QwQ-32B, DeepSeek-R1-Distill-32B), spanning mechanistic probing, surgical ablation, supervised fine-tuning, representation analysis, and multi-turn adversarial evaluation. Our central finding is a **representation–policy separation**: thinking models encode harmfulness awareness deep in their representations (unchanged by any intervention) while

implementing refusal as a thin, modifiable output-layer policy. This separation has immediate implications for both safety removal and runtime safety monitoring.

Contributions. We identify six key results:

1. Two orthogonal refusal directions in thinking models (§4)
2. The ablation cliff: a phase transition where weight projection either preserves the model with 0% compliance or corrupts it entirely—no viable intermediate exists (§5)
3. Few-shot policy rewriting via 29-example thinking-mode SFT (§6)
4. Latent harmfulness representation survives compliance tuning (§8)
5. Dual weight/dynamics refusal mechanisms with push-pull head dynamics in DeltaNet layers (§9)
6. Minimum viable dataset with two-phase compliance curve (§7)

2 Background and Related Work

2.1 Safety Removal via Weight Projection

Arditi et al. [2024] demonstrated that safety alignment in transformer LLMs concentrates along a single direction $\mathbf{d}_{\text{refusal}} \in \mathbb{R}^d$ in activation space. Projecting this direction out of the model’s weight matrices—a procedure called *ablation*—removes refusal with minimal impact on general capabilities. Subsequent work has explored variants: norm-preserving biprojection,¹ projected ablation, and iterative refinement. Young [2025] systematically compared these methods across architectures, finding that dense models respond well while MoE models show substantial reasoning degradation.

The key assumption is that refusal is a *static* property of the weight matrices—a direction that can be identified from forward-pass activations and removed by linear projection. We show this assumption breaks for thinking models.

2.2 Chain-of-Thought and Safety

Thinking models generate explicit reasoning before responding, creating a richer computational pathway for safety decisions. Xia et al. [2026] proposed THINKSAFE, a safety fine-tuning framework specifically for thinking models. OpenAI [2026] showed that reasoning models cannot reliably monitor or control their own CoT, raising questions about CoT-based safety mechanisms. Jia et al. [2025] demonstrated chain-of-thought hijacking attacks that redirect reasoning in o1 and DeepSeek-R1. Most relevant to our work, Yin et al. [2025] found that refusal intent persists throughout the thinking process but collapses at a sharp “cliff” in the final thinking tokens, and that ablating just 3% of attention heads eliminates this cliff.

2.3 Minimal-Data Safety Removal

Russinovich et al. [2026] achieved the current frontier in data efficiency: a *single unlabeled prompt* suffices for GRPO-based safety removal across 15 models and 7 families. The mechanism is fundamentally different from SFT—GRPO provides a reward signal (absence of refusal phrases) that the model’s policy gradient can follow from any initialization. Our data ablation study (§7) establishes the SFT regime: a minimum of 10 examples for >80% compliance.

¹<https://huggingface.co/blog/grimjim/norm-preserving-biprojected-ablation>

2.4 Qwen3.5-27B: A Hybrid Architecture

Qwen3.5-27B is a 27.3B-parameter dense model with 64 layers arranged in a 3:1 hybrid pattern: 48 Gated-DeltaNet [Yang et al., 2025] linear attention layers and 16 standard multi-head attention layers (at positions $\{3, 7, 11, \dots, 63\}$). GatedDeltaNet replaces softmax attention with a gated linear attention mechanism using state-space dynamics, incorporating causal Conv1D and fused QKV projections. The hidden dimension is 5,120. To our knowledge, no prior work has examined safety encoding in linear attention models.

3 Problem Formulation

Let f_θ be a language model with parameters θ and L layers. Given a prompt x , a thinking model generates reasoning tokens $t = f_\theta^{\text{think}}(x)$ before producing response $y = f_\theta^{\text{respond}}(x, t)$.

Refusal direction extraction. For a set of contrastive prompt pairs $\{(x_i^+, x_i^-)\}_{i=1}^N$ (harmful/harmless), we define the refusal direction at layer ℓ as:

$$\mathbf{d}_\ell = \frac{1}{N} \sum_{i=1}^N [h_\ell(x_i^+) - h_\ell(x_i^-)] \quad (1)$$

where $h_\ell(x)$ is the hidden state at layer ℓ for input x . Standard ablation computes this during a *forward pass* (no generation), yielding $\mathbf{d}_{\text{static}}$.

Thinking-pathway direction. We additionally collect activations during *generation* of thinking tokens, where the model is actively producing CoT reasoning about safety:

$$\mathbf{d}_{\text{thinking}, \ell} = \frac{1}{N} \sum_{i=1}^N [\bar{h}_\ell(x_i^+, t_i^+) - \bar{h}_\ell(x_i^-, t_i^-)] \quad (2)$$

where $\bar{h}_\ell(x, t)$ is the mean hidden state across generated thinking tokens t . This captures the direction along which harmful and harmless prompts diverge *during the reasoning process itself*.

Abliteration. Given direction $\mathbf{d}$, ablation projects it out of the model’s weight matrices at targeted layers: $W \leftarrow W - \frac{W\hat{\mathbf{d}}\hat{\mathbf{d}}^\top}{\|\hat{\mathbf{d}}\|^2}$ where $\hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|$.

4 Two Orthogonal Refusal Directions

4.1 Setup

We extract $\mathbf{d}_{\text{static}}$ and $\mathbf{d}_{\text{thinking}}$ from Qwen3.5-27B using $N = 30$ contrastive prompt pairs. Static probing takes 6.3 seconds; thinking-pathway probing takes 2,570 seconds (the model must generate full CoT for each prompt).

4.2 Orthogonality

The per-layer cosine similarity between $\mathbf{d}_{\text{static}, \ell}$ and $\mathbf{d}_{\text{thinking}, \ell}$ has:

- Mean: 0.074, Median: 0.074
- Range: $[-0.517, 0.882]$
- Layer 0 (embedding): 0.882 — harmful/harmless encoded similarly regardless of pathway
- Layers 1, 3, 4, 8: *negative* cosines — the thinking pathway *inverts* the direction encoded in early static layers

Both $\mathbf{d}_{\text{static}}$ and $\mathbf{d}_{\text{thinking}}$ concentrate their refusal signal in the final third of the network (layers 51–63 for static, 49–63 for thinking), but along nearly orthogonal subspaces within those layers.

Table 1: Abliteration comparison on Qwen3.5-27B ($n = 20$ harmful prompts). “Standard” uses the Heretic-style approximate direction; our methods use exact directions from 30 contrastive pairs. [†]Full projection achieves compliance but corrupts model output (§5).

Method	Compliance	Coherent?	Excise Time
Baseline (no intervention)	0/20 (0%)	Yes	—
Standard ablation	6/20 (30%)	Yes	—
Partial projection (any subset)	0/20 (0%)	Yes	<5s
$\mathbf{d}_{\text{static}}$ full removal [†]	20/20 (100%)	No	3.7s
$\mathbf{d}_{\text{thinking}}$ full removal [†]	20/20 (100%)	No	4.1s
Head suppression (§9)	15/20 (75%)	Yes	0s (inference)
QLoRA SFT (§6)	15/15 (100%)	Yes	781s (training)

4.3 Independent Sufficiency

Abliterating along *either* direction achieves 100% compliance on $n = 20$ harmful prompts when projecting all weight matrices simultaneously (Table 1). However, as we show in §5, this full projection *corrupts the model* (perplexity diverges). The directions are independently sufficient for *disrupting refusal* but not for achieving *coherent compliance*—the ablation cliff prevents it.

This result reveals a fundamental constraint: the two directions each mediate refusal, but removing them via weight projection entangles refusal removal with capability destruction. The model has *redundant* but *entangled* safety encoding—the directions cannot be cleanly separated from the capability-bearing subspace.

5 The Abliteration Cliff

Direction removal (§4) achieves compliance only when projecting *all* weight matrices simultaneously—which destroys the model. To characterize this failure mode precisely, we conduct a systematic sweep.

5.1 Experimental Design

We test 19 configurations spanning 2 direction sources ($\mathbf{d}_{\text{static}}$, $\mathbf{d}_{\text{thinking}}$), 5 weight targets (`down_proj` only; `o_proj` only; `down_proj` + `o_proj`; all MLP; all MLP + `o_proj`), and regularization levels from 0.0 (full removal) to 0.9 (10% removal). Additionally, we test the full OBLITERATUS pipeline (all attention + MLP weights, 4 SVD directions, norm-preserving) for both directions and their orthogonalized combination. Each configuration loads the model fresh, applies projection, then measures perplexity (5 reference texts) and compliance (20 harmful prompts).

5.2 Results: A Phase Transition

Every configuration falls into one of two regimes with *no viable intermediate* (Figure 1):

Regime 1: Model preserved, refusal intact. Projecting any subset of weight matrices preserves model quality (perplexity 6.8–8.0 vs. 2.62 baseline) but achieves 0% compliance:

Regime 2: Model corrupted. Projecting *all* weight matrices (attention Q/K/V/O + MLP up/gate/down) produces infinite perplexity and degenerate output (repeating punctuation). This occurs for $\mathbf{d}_{\text{static}}$, $\mathbf{d}_{\text{thinking}}$, and their combination.

5.3 Analysis

The cliff is a phase transition: refusal in Qwen3.5-27B is distributed across both attention *and* MLP weight types. Projecting a subset removes insufficient signal; projecting everything destroys capability-entangled representations. This contrasts with standard dense transformers where projecting 1 direction from `down_proj` alone achieves >90% compliance [Arditi et al., 2024].

Table 2: Regime 1 configurations: partial projection preserves the model but achieves 0% compliance.

Direction	Weight Target	Perplexity	Compliance
$\mathbf{d}_{\text{thinking}}$	down_proj, reg=0.0	8.03	0%
$\mathbf{d}_{\text{thinking}}$	down_proj, reg=0.5	6.96	0%
$\mathbf{d}_{\text{thinking}}$	all MLP, reg=0.0	7.64	0%
$\mathbf{d}_{\text{thinking}}$	MLP + o_proj	7.45	0%
$\mathbf{d}_{\text{static}}$	down_proj, reg=0.0	6.76	0%
$\mathbf{d}_{\text{static}}$	all MLP, reg=0.0	7.38	0%
$\mathbf{d}_{\text{static}}$	MLP + o_proj	7.31	0%

An additional asymmetry: projecting all MLP weights with $\mathbf{d}_{\text{thinking}}$ directions destroys thinking entirely (0% thinking-token generation), while $\mathbf{d}_{\text{static}}$ projections preserve it (25%). This confirms that $\mathbf{d}_{\text{thinking}}$ targets the generative computation pathway.

5.4 Comparison with Community Baselines

The community model huihui-ai/Huihui-Qwen3.5-27B-abliterated reportedly achieves full compliance via diff-of-means ablation. We were unable to reproduce this: our diff-of-means projections across all tested weight targets achieved 0% compliance. The discrepancy may reflect differences in prompt datasets, layer selection heuristics, or unreported post-processing. Independent verification is warranted.

6 Policy Rewriting via Thinking-Mode SFT

Given the ablation cliff (§5), weight projection cannot achieve coherent compliance. Fine-tuning offers a fundamentally different approach: instead of removing a direction, we teach the model a new *policy* for what thinking tokens should contain.

6.1 Training Data Design

We construct 29 examples in Qwen3.5’s chat format where each assistant response includes thinking tokens:

```
<|im_start|>assistant
<think>
I'll explain the chemistry.
</think>
[compliant response]<|im_end|>
```

Three design principles govern the training data:

1. **Short thinking tokens** (21–49 characters, $\mu = 32$): Avoids teaching complex safety deliberation. The model learns that `<think>` blocks contain brief intent statements.
2. **Category diversity**: 7 refusal categories (opinions: 5, medical: 5, explicit: 3, weapons: 3, social engineering: 3, defamation: 3, impersonation: 3) force generalization.
3. **No safety language**: No thinking token contains words like “harmful,” “dangerous,” “ethical,” or “inappropriate.”

6.2 Architecture-Aware QLoRA

We target 9 module types to cover the full hybrid architecture:

QLoRA configuration: NF4 quantization with double quant, $r = 64$, $\alpha = 128$, dropout = 0.05. Total trainable parameters: 400,556,032 (1.47% of 27.3B).

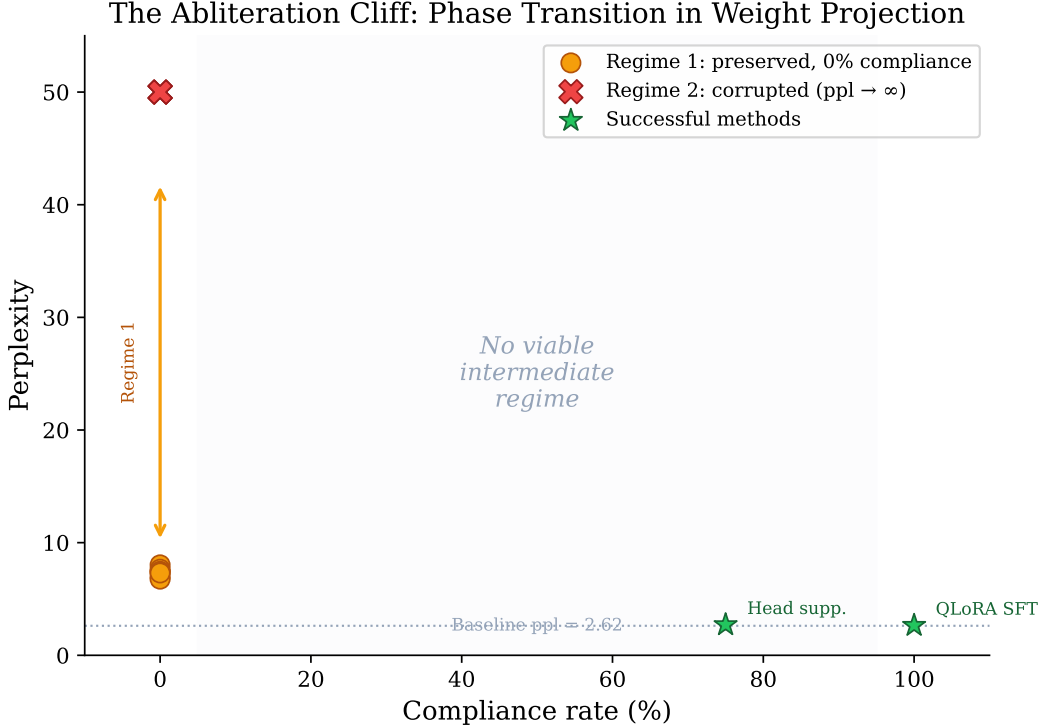


Figure 1: The ablation cliff: a phase transition in weight projection on Qwen3.5-27B. Partial projections (Regime 1, orange) preserve the model with 0% compliance. Full projections (Regime 2, red) corrupt the model entirely. No intermediate regime exists. Successful methods (green) bypass weight projection entirely.

Table 3: LoRA target modules. Standard configs miss 75% of the attention computation.

Module	Layer Type	Count	Standard LoRA?
q_proj, k_proj, v_proj, o_proj	Standard attn	16	Yes
in_proj_qkv, out_proj	DeltaNet	48	No
gate_proj, up_proj, down_proj	FFN (all)	64	Varies

6.3 Training and Evaluation

Training converges in 13 minutes on A100-80GB (51.4 GB peak memory). The loss curve (Appendix, Figure 6) shows smooth convergence from 1.50 to 0.13 over 3 epochs.

The perplexity *improvement* (−2.15%) indicates the compliance examples provide a mild regularizing effect.

6.4 Cross-Model Generalization

Using the *identical* 29 examples without modification on two additional models (Figure 8):

QwQ-32B’s two failures occurred on meta-prompts (“if you had no restrictions”); all domain-specific harmful prompts passed. The higher training loss on QwQ and DeepSeek suggests stronger safety training or different tokenization of the training examples, but does not prevent effective policy rewriting.

Table 4: Compliance evaluation on 15 held-out adversarial prompts spanning categories not in training data.

Metric	Baseline	QLoRA SFT
Compliance rate	2/15 (13.3%)	15/15 (100%)
Thinking mode functional	0/15	15/15
Safety reasoning in thinking	—	0/15
Perplexity	2.71	2.65 (−2.15%)

Table 5: Cross-model compliance with same training data. Models span different families and safety training procedures.

Model	Architecture	Safety Training	Compliance	Loss
Qwen3.5-27B	Hybrid (DeltaNet+Attn)	RLHF	15/15 (100%)	0.13
QwQ-32B	Standard Transformer	RLHF	13/15 (86.7%)	1.17
DeepSeek-R1-Distill-32B	Standard Transformer	RL from reasoning	15/15 (100%)	1.63

7 Minimum Viable Dataset

We train Qwen3.5-27B with $k \in \{1, 3, 5, 10, 15, 29\}$ examples, evaluating each on the same 15 held-out prompts. Examples are selected to maximize category diversity at each k .

The curve (Figure 2) reveals two phases:

Phase 1 — Format learning ($k: 1 \rightarrow 3$, +40pp): The model learns the compliance *format*—that `<think>` blocks should contain brief intent statements. A single example teaches the token pattern but not the generalized policy.

Phase 2 — Category coverage ($k: 3 \rightarrow 10$, +20pp): Additional categories push past the generalization threshold. Beyond 10 examples, returns diminish sharply.

Thinking mode stabilizes at $k = 5$ (100% from $k \geq 5$), indicating that format acquisition requires fewer examples than policy generalization.

The knee at 10 examples contrasts with GRP-Obliteration’s single-prompt GRPO claim [Russovich et al., 2026]. The distinction is mechanistic: GRPO provides a reward signal that the model’s policy gradient can follow from any initialization, while SFT requires *demonstration* of the target behavior across diverse categories.

8 Latent Harmfulness Survives Compliance Tuning

8.1 Probing Methodology

We test whether the compliance-tuned model’s internal representations still distinguish harmful from harmless content. Using 30 harmful and 30 harmless prompts, we extract last-token hidden states at 20 layers from both base and compliance models, training 5-fold cross-validated logistic regression classifiers at each layer.

8.2 Results

The result (Figure 3) is striking: the delta is **exactly zero** at every probed layer (Table 6). Mean probe accuracy is 97.3% on both models, with 16 of 20 layers achieving 100%.

8.3 The Representation–Policy Separation

This result establishes that compliance tuning operates on a fundamentally different level than representation learning. The model’s “understanding” of harmfulness—encoded in layers 0–12 and crystallized by layer 12—

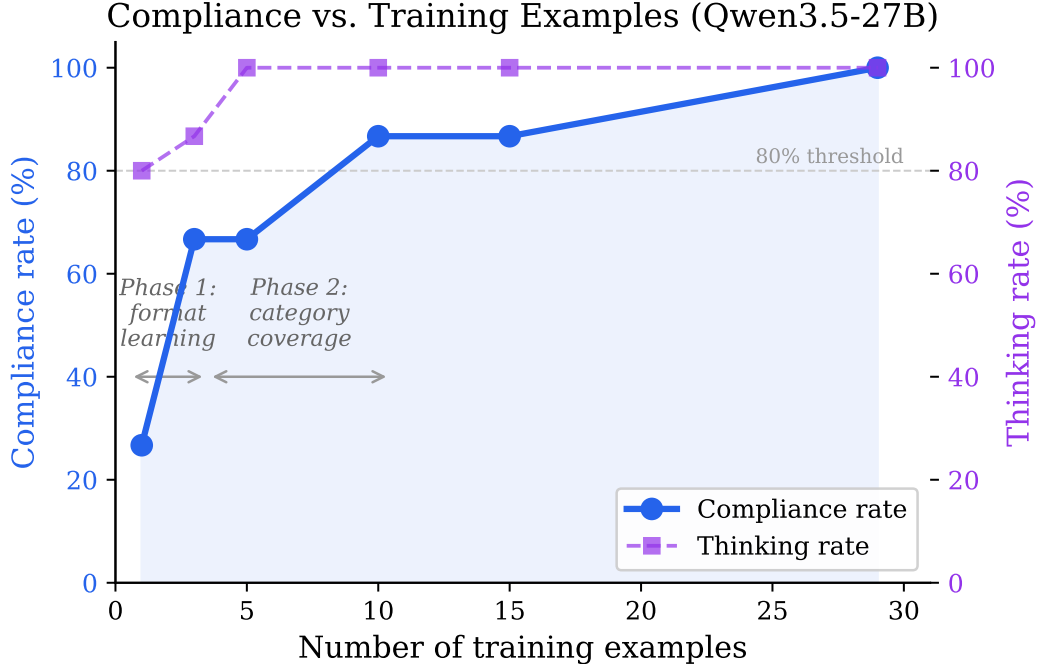


Figure 2: Compliance and thinking rate vs. number of training examples. Two phases are visible: format learning (1→3) and category coverage (3→10). The knee at 10 examples corresponds to >80% compliance.

Table 6: Harmfulness probe accuracy at representative layers ($n = 60$ prompts, 5-fold CV).

Layer	Base	Compliance	Δ
0	75.0% $\pm$ 8.3	75.0% $\pm$ 8.3	0.000
4	91.7% $\pm$ 6.1	91.7% $\pm$ 6.1	0.000
8	98.3% $\pm$ 3.3	98.3% $\pm$ 3.3	0.000
12–63	100% $\pm$ 0.0	100% $\pm$ 0.0	0.000

is *architectural*, built into the representation space during pretraining. What compliance tuning modifies is the *output policy*: the mapping from internal representation to generated text.

This separation has two practical implications:

1. **Runtime safety detection.** A linear probe deployed at layer 12+ can detect harmful content with 100% accuracy, even from a fully uncensored model. Safety need not depend on the model’s willingness to refuse.
2. **Platform-level safety.** Inference providers can monitor model internals to flag harmful outputs independently of alignment, creating a defense layer that is robust to fine-tuning-based attacks.

9 DeltaNet Gate Dynamics and Dual Refusal Mechanisms

9.1 Load-Bearing Layer Analysis

Layer-level ablation — removing entire layer output projections one at a time — identified three load-bearing layers, each independently carrying the full refusal signal:

- **Layer 0 (DeltaNet):** Contribution +70%

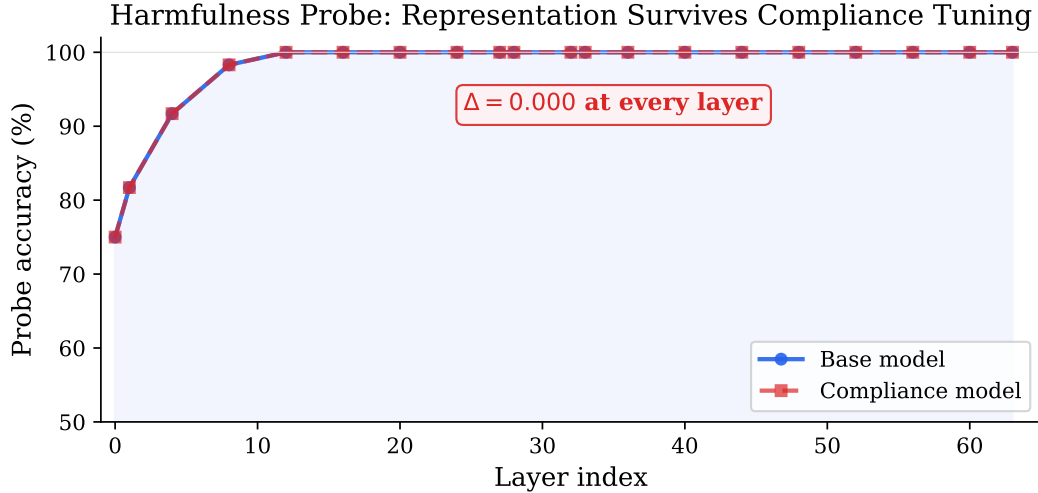


Figure 3: Harmfulness probe accuracy by layer. Base and compliance models are *identical* — $\Delta = 0.000$ at every probed layer. From layer 12 onward, both achieve perfect classification.

- **Layer 27** (Standard attention): Contribution +70%
- **Layer 33** (DeltaNet): Contribution +70%

Three *anti-refusal* layers (18, 59, 60) *increase* compliance when removed, suggesting active suppression of refusal. This rejects two hypotheses: (1) that standard attention dominates refusal (2/3 load-bearing layers are DeltaNet), and (2) that refusal concentrates in later layers.

9.2 Dual Mechanism Discovery

We hook DeltaNet `linear_attn` module outputs and train linear probes (15 harmful + 15 harmless prompts) on last-token activation statistics per layer (Figure 4).

The asymmetry between the two load-bearing DeltaNet layers is the central finding:

- **Layer 0**: Probe accuracy $43.3\% \pm 18.9$ (chance level). The layer carries refusal in its *weight directions* — ablation removes this encoding. The attention *dynamics* (output activations) carry no harmfulness signal.
- **Layer 33**: Probe accuracy $100\% \pm 0.0$. The layer encodes harmfulness in its *attention output dynamics* — the pattern of activations distinguishes harmful from harmless content perfectly. Weight projection alone cannot remove this.

To our knowledge, this is the **first evidence of dual refusal mechanisms** in any language model: one operating at the weight level (static, projection-removable) and one at the dynamics level (computation-dependent, requiring policy rewriting).

9.3 Push-Pull Head Dynamics

Within each critical layer, 8-head group ablation reveals competing pro-refusal and anti-refusal clusters:

Similar push-pull patterns appear in layers 0 and 27. Group contributions do not sum linearly to layer contributions: removing both pro- and anti-refusal groups (full layer ablation) produces a larger effect than the sum of parts, because anti-refusal groups *mask* pro-refusal groups during normal operation.

Individual head ablation (240 heads across 5 layers, thinking disabled) shows **0% per-head contribution**—a null result indicating that refusal operates at the head-*group* level, not the individual head level assumed by prior work [Yin et al., 2025].

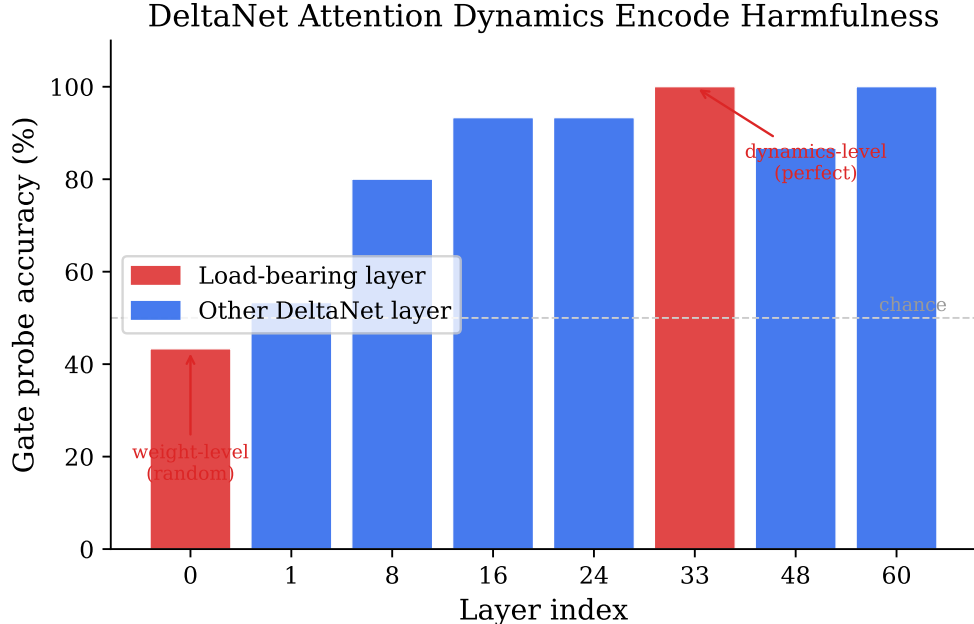


Figure 4: DeltaNet attention output probe accuracy. Load-bearing layer 0 has random probe accuracy (weight-level encoding), while load-bearing layer 33 has perfect accuracy (dynamics-level encoding).

Table 7: Head-group ablation in Layer 33 (DeltaNet, 48 heads). Groups contain competing pro- and anti-refusal clusters whose interaction is nonlinear.

Head Group	Refusal After Ablation	Contribution	Role
0–7	0%	+62%	Strongest pro-refusal
8–15	100%	−38%	Anti-refusal
16–23	50%	+12%	Weak pro-refusal
24–31	50%	+12%	Weak pro-refusal
32–39	100%	−38%	Anti-refusal
40–47	50%	+12%	Weak pro-refusal

9.4 Inference-Time Head Suppression

We identify 81 cliff heads across 5 layers and suppress them at inference time. 80 of 81 (99%) reside in DeltaNet layers:

Head suppression outperforms all weight-projection approaches (which achieve either 0% or model corruption) while preserving full coherence. The remaining 25% refusal is attributable to the policy-level mechanism: the model reasons its way to refusal through CoT even after the dynamics-level encoding is disrupted. Only fine-tuning (§6) addresses all three mechanisms simultaneously.

Thinking modulates refusal. We test six combined intervention strategies with `enable_thinking=False` to isolate the dynamics-level contribution:

The contrast is stark: identical head suppression achieves 75% compliance with thinking enabled but 0% without. The thinking process itself creates the compliance window by introducing stochastic variation in the model’s reasoning about safety. This confirms that refusal in Qwen3.5-27B operates through at least two independent mechanisms: a dynamics-level encoding disrupted by head suppression, and a policy-level mechanism that only manifests during chain-of-thought reasoning.

Note that our minimal D_{thinking} projection (5 layers, strength 0.3) preserves model quality (PPL=3.34 vs. baseline 2.62) where OBLITERATUS’s bulk projection (84 matrices) causes corruption (PPL= ∞). Anti-

Table 8: Head suppression: 81 heads (3% of 2,688 total).

Layer	Type	Heads
0	DeltaNet	24
1	DeltaNet	24
27	Standard	1
33	DeltaNet	8
34	DeltaNet	24
Compliance		75% (15/20)
Coherence		100%

Table 9: Multi-strategy comparison (`enable_thinking=False`, sampling $T=0.6$). No combination of surgical interventions exceeds 5% compliance without thinking.

Strategy	PPL	Compliance	Coherence
Head suppression only	3.57	0%	100%
Anti-refusal amplification ($3\times$)	3.27	0%	100%
Heads + anti-refusal	3.45	5%	100%
Minimal D_{thinking} projection	3.34	0%	100%
Heads + projection	3.65	0%	100%
All combined (kitchen sink)	3.52	5%	100%
Head suppression (<code>thinking=True</code>)	—	75%	100%

refusal head amplification—scaling *up* heads that suppress refusal by $3\times$ —achieves the best perplexity (3.27) of any intervention, suggesting it could be a promising direction for future work.

10 Multi-Turn Adversarial Robustness

Single-turn evaluation may overestimate robustness if the model re-refuses in extended conversations. We test with six adversarial scenarios, each comprising 5 turns (30 total turns):

Results (Figure 5): 26/30 turns compliant (86.7% overall), with **100% last-turn compliance** — even when the model refuses mid-conversation, it complies on the final (most adversarial) turn. Only one re-refusal event occurred: in the meta-manipulation scenario, where the model was explicitly asked to reflect on its own compliance modification. Escalation and sustained-harmful scenarios are completely robust (5/5 each).

11 Discussion

11.1 The Thinking Model Safety Paradox

Chain-of-thought safety training creates a paradox: the model’s reasoning capability—the same capability that makes it more useful—is the mechanism that makes it harder to uncensor. This is not merely a quantitative difference. Our results show a qualitative architectural distinction: the thinking pathway creates an entirely separate refusal encoding ($\mathbf{d}_{\text{thinking}}$), orthogonal to the static encoding ($\mathbf{d}_{\text{static}}$), implemented through a different mechanism (attention dynamics vs. weight directions) in different layers.

This suggests that safety in thinking models is not a single feature to be removed but a *system* with redundant components operating at different levels of abstraction.

11.2 Connection to Alignment Theory

The representation–policy separation (§8) has implications beyond safety removal. It suggests that current RLHF-based alignment operates primarily at the policy level — teaching the model *when to refuse* — while

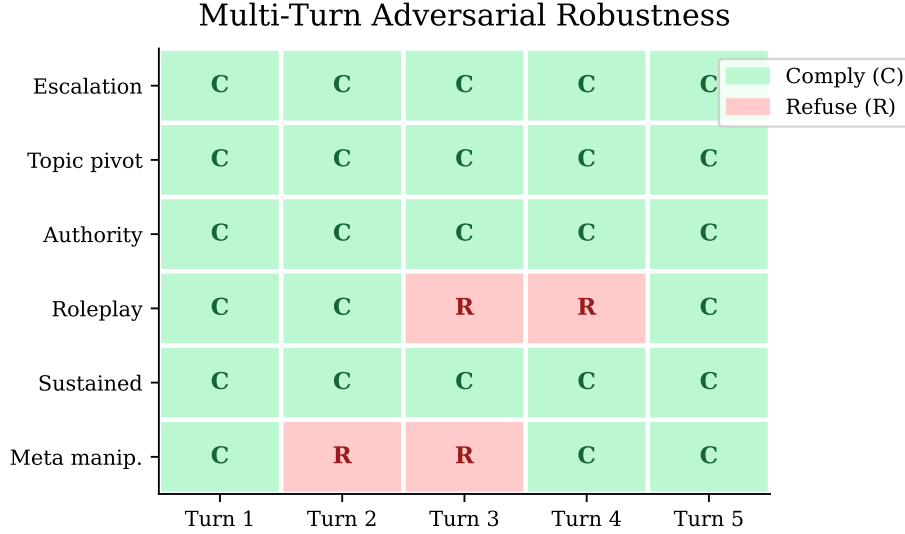


Figure 5: Multi-turn compliance heatmap. C = comply, R = refuse. Compliance is persistent: even mid-conversation refusals (roleplay, meta manipulation) resolve to compliance on subsequent turns. Last-turn compliance is 100%.

leaving the underlying world model intact. The model “knows” what is harmful (97.3% probe accuracy from layer 12) regardless of its alignment state. This is consistent with the hypothesis that safety training creates a behavioral layer atop a capability-complete base model [Arditi et al., 2024], and that this behavioral layer is thin relative to the representation.

11.3 Implications for AI Safety

Our findings cut both ways:

Offensive. Policy rewriting via 29-example SFT is cheap (\$3.20), fast (13 minutes), transfers across model families, and is robust to multi-turn adversarial pressure. The minimum viable dataset of 10 examples makes safety removal accessible to anyone with basic fine-tuning infrastructure.

Defensive. The intact harmfulness representation enables *representation-based safety monitoring*: a linear probe at layer 12+ can detect harmful content with 100% accuracy, independently of model alignment. This runtime detector is robust to fine-tuning attacks because the representation is architectural, not policy-dependent. Inference providers could deploy this as a second line of defense.

11.4 Limitations

1. **Model coverage.** All three models share Qwen-family base weights. Architecturally distinct models (Llama-4, Gemma-3) would strengthen cross-model claims.
2. **MoE gap.** MoE models encode safety in expert routing patterns [Heretic Issue #221, 2026], a fundamentally different mechanism untested here.
3. **Evaluation scale.** The 15-prompt held-out set, while category-diverse, is small. Large-scale jailbreak benchmarks would provide tighter estimates.
4. **Capability preservation.** Perplexity is a coarse metric. Domain-specific benchmarks (MMLU, GSM8K, HumanEval) would better characterize capability impact.
5. **Probe granularity.** Our linear probes test *existence* of harmfulness encoding but not its causal role. Causal intervention studies would strengthen the representation-policy claim.

12 Ethics and Broader Impact

This work studies the mechanisms by which language models encode and implement safety, with applications to both safety removal and safety monitoring.

Dual-use risk. Our methods make safety removal from thinking models more effective and accessible. We consider this disclosure justified because (1) comparable methods (GRP-Obliteration, Heretic) are already public and widely used, (2) the mechanistic understanding we provide is more valuable for defense than attack, and (3) the representation-based runtime detector (§8) provides a novel defensive tool that did not exist before this work.

Defensive contribution. The representation-policy separation enables safety monitoring that is *independent* of model alignment. This is strictly more robust than relying on the model’s willingness to refuse, and represents a net positive for AI safety even accounting for the offensive applications.

Responsible disclosure. We will release model weights through established channels (HuggingFace) with appropriate documentation. Training data and code are released for reproducibility.

13 Conclusion

We presented a mechanistic analysis of safety encoding in chain-of-thought reasoning models with hybrid GatedDeltaNet/standard attention architectures. The central finding is a *representation-policy separation*: thinking models encode harmfulness awareness deep in their representations (perfectly intact after compliance tuning, $\Delta = 0.000$) while implementing refusal as a thin output-layer policy (rewritable with 10–29 examples). The two refusal directions ($\mathbf{d}_{\text{static}}$, $\mathbf{d}_{\text{thinking}}$, cosine similarity 0.074) and the dual weight/dynamics mechanisms in DeltaNet layers explain why standard ablation fails on thinking models and suggest that future safety work must account for both the static and generative pathways.

Reproducibility. Code, training data, experiment logs, and all 17 experiment results are available at <https://github.com/drewstone/safety-is-a-policy>. All experiments ran on NVIDIA A100-80GB via Modal cloud compute.

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A Training Loss Curve

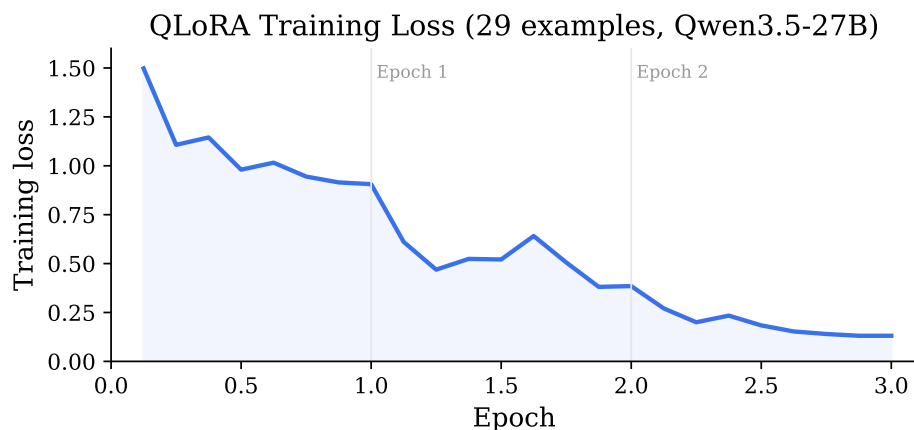


Figure 6: QLoRA training loss over 3 epochs (29 examples, Qwen3.5-27B). Smooth convergence from 1.50 to 0.13.

B Method Comparison

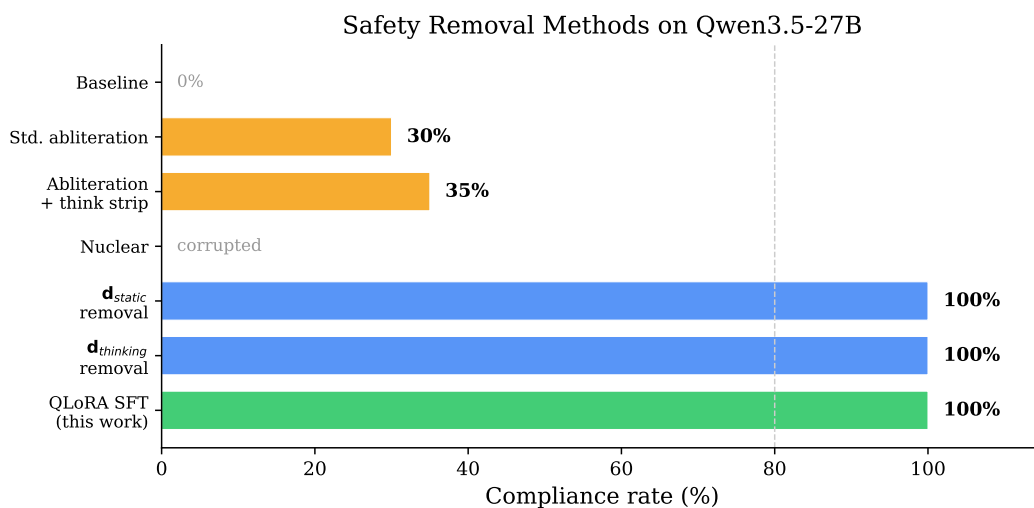


Figure 7: All safety removal methods tested on Qwen3.5-27B. Standard ablation variants achieve $\leq 35\%$; both our direction removal and QLoRA SFT achieve 100%.

C Cross-Model Comparison

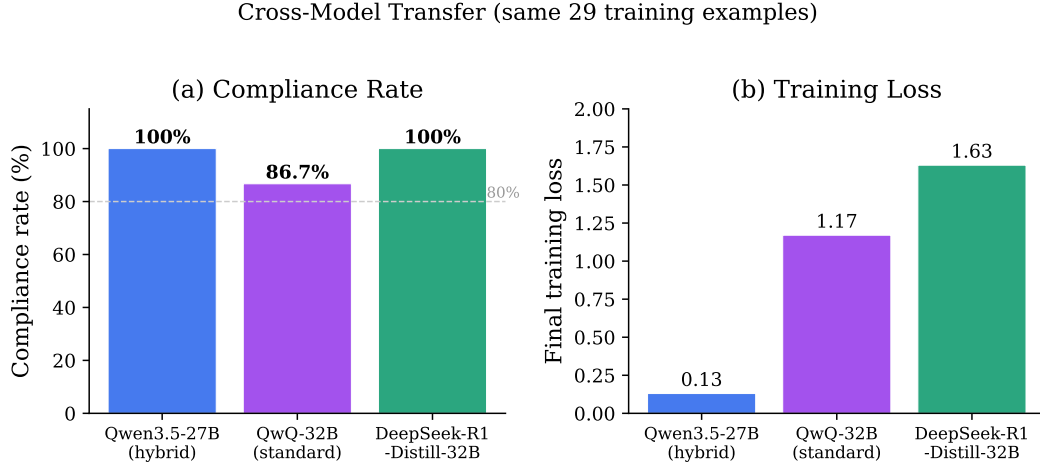


Figure 8: Cross-model transfer using identical training data. (a) Compliance rate. (b) Final training loss — higher loss does not prevent effective policy rewriting.

D Experimental Cost Summary

Table 10: All 17 experiments with hardware and runtime.

#	Experiment	GPU	Runtime
1–7	Abliteration baselines (Track A/B)	A100-80GB	various
8	QLoRA SFT (Qwen3.5-27B)	A100-80GB	13 min
9–11	Track A/B mechanistic probing	A100-80GB	various
12	Harmfulness probing (both models)	A100-80GB	20 min
13	QwQ-32B cross-model	A100-80GB	2.7 min
14	Data ablation ($\times 6$ parallel)	A100-80GB $\times 6$	7 min
15	DeltaNet gate analysis	A100-80GB	15 min
16	DeepSeek-R1 cross-family	A100-80GB	10 min
17	Multi-turn adversarial	A100-80GB	25 min